

STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation

Details	Information
Student Name	Anjali Shikhare
Course	Python with AI
Project Type	Python Data Analysis Project
Technologies	Python, NumPy, Pandas
Dataset	20 student records
Submission Year	2026

This document explains the objective, dataset, implementation, features, results, learning outcomes and final outcome of the Student Performance Analytics System.

1. Objective

The aim of this project is to analyze student marks and attendance using Python, NumPy and Pandas. The system reads student data from a CSV file, processes the marks, calculates performance measures and presents useful results such as total marks, average marks, grades, pass/fail status, ranking, subject-wise performance and top performers.

2. Technologies Used

Technology	Purpose
Python	Used for the main program, functions, conditions and loops.
NumPy	Used for numerical calculations such as sum, mean, median and standard deviation.
Pandas	Used to read the CSV file, create/process the DataFrame and perform data analysis.
CSV	Used to store the student dataset.

3. Dataset Description

The project uses a CSV file named students.csv containing 20 student records. The dataset includes student identification, department, marks in five subjects and attendance.

Field	Description
Student_ID	Unique ID of the student.
Name	Student name.
Department	Student department such as CSE, IT, ECE, Civil and Mechanical.
Python	Marks in Python.
Java	Marks in Java.
DBMS	Marks in DBMS.
Web Development	Marks in Web Development.
Data Analytics	Marks in Data Analytics.
Attendance	Attendance percentage.

4. Implementation

Step 1 – Load the Dataset

The `load_data()` function uses Pandas `read_csv()` to read `students.csv` and return the student data as a `DataFrame`.

Step 2 – Calculate Total and Average

For each student, the five subject marks are selected. NumPy is used to calculate the total using `np.sum()` and the average using `np.mean()`.

Step 3 – Grade and Result

Grades are assigned according to the calculated average. The grading rule used in the project is: A+ for 90 and above, A for 80–89, B+ for 70–79, B for 60–69, C for 50–59, D for 40–49 and F below 40. A student is marked Fail if any subject mark is below 40; otherwise the result is Pass.

Step 4 – Attendance Status

Attendance is checked using a minimum attendance value of 75%. Students with attendance of 75% or above are marked Regular; otherwise they are marked Low.

Step 5 – Ranking and Analysis

Students are ranked according to total marks. The project also calculates class average, median, standard deviation, pass percentage and average attendance. Subject-wise analysis finds average, highest, lowest, passed, failed and subject topper values.

5. Project Structure

File	Purpose
<code>main.py</code>	Runs the project and displays the final analysis.
<code>functions.py</code>	Contains reusable functions for calculations and analysis.
<code>students.csv</code>	Stores the 20 student records.
<code>README.md</code>	Provides basic project information and running instructions.

6. Key Features

- Reads and displays student data from a CSV file.
- Calculates total marks for each student.
- Calculates average marks using NumPy.
- Assigns grades based on the average.
- Identifies Pass/Fail based on subject marks.
- Ranks students based on total marks.
- Performs subject-wise analysis.
- Identifies the top five students.
- Checks attendance status using the 75% threshold.
- Calculates class-level performance statistics.

7. Output / Results

The program was executed successfully using the project files. The following results were obtained from the current 20-student dataset.

Metric	Result
Total Students	20
Class Average	71.88
Highest Average	Ananya Bhosale – 93.2
Lowest Average	Rahul Sawant – 44.0
Students Passed	17
Students Failed	3
Pass Percentage	85.0%

Subject-wise Analysis

Subject	Average	Highest	Lowest	Topper
Python	70.70	95	35	Priya Deshmukh
Java	69.20	95	38	Ananya Bhosale
DBMS	70.20	92	39	Ananya Bhosale
Web Development	74.75	92	50	Neha Pawar
Data Analytics	74.55	98	46	Priya Deshmukh

8. Top Performers

Rank	Student	Department	Total	Average	Grade
1	Ananya Bhosale	ECE	466	93.2	A+
2	Priya Deshmukh	CSE	460	92.0	A+
3	Neha Pawar	IT	454	90.8	A+
4	Meera Phadke	Civil	437	87.4	A
5	Ishita Kamble	CSE	424	84.8	A

Sample Output

The console output displays sections for Student Data, Results, Overall Performance, Subject Analysis and Top 5 Students. The program finishes with “PROJECT COMPLETED”.

Example: Total Students: 20 | Class Average: 71.88 | Students Passed: 17 | Students Failed: 3 | Pass Percentage: 85.0%

9. Final Outcome

The Student Performance Analytics System successfully reads student data and performs basic academic analysis using Python, NumPy and Pandas. It provides student-wise results, grades, pass/fail status and rankings, along with subject-wise and class-level performance information. The project demonstrates the use of Python fundamentals, NumPy calculations and Pandas DataFrame operations in a practical data-analysis application.

10. Challenges and Learning

While developing the project, attention was required to keep the subject names consistent between the Python code and CSV dataset. The project also helped in understanding how to divide calculations into reusable functions and how to use loops, conditions, NumPy and Pandas together for a complete analysis task.

11. GitHub Repository

GitHub repository: <https://github.com/anjalishikhare80/student-performance-analytics>

12. Conclusion

This project meets the main requirements of the Student Performance Analytics System assignment. It uses a 20-record student dataset, Pandas for data handling, NumPy for numerical calculations, functions and loops for processing, and produces meaningful academic performance results.

Final Project Files

File	Status
main.py	Completed and tested
functions.py	Completed and tested
students.csv	20 student records
README.md	Project information and run instructions
Project_Documentation.docx	This documentation

End of Project Documentation